6 Round eleven: the decoder takes the rules' functional form ✗

What the round asked. Ten rounds left three readings: composition is what buys the private readout, no public screen lifts the target or is learned at a held-out compound, and every dose arm over-predicts the isobole interiors. Round eleven tested five new parts of the model, each switchable and each defaulting to the round-ten arm, so every cell of the round is an ablation of one part. *Displacement*: the compound token moves every gene embedding along a learned direction, gated per gene, which is the environment's version of the encoder's deletion operator, against the round-ten design in which the compound tokens attend to the genes. *Cell-only*: the head reads the perturbed cell state and nothing else, so a compound's whole effect has to pass through the gene embeddings. *Hill form*: a second readout of the same state gives each inhibitor a dose-response curve, an IC50 and a slope, and the host's fitness is that curve at the medium's dose. *Composition*: the single-inhibitor readings of a mixture combine by Bliss independence, by Loewe additivity solved on the curves, or by the highest single agent, the three model-free rules of the figure now computed inside the decoder. *Pair term*: a learned correction on every pair of a mixture's compounds, zero at the start, which is the environment's analog of a genetic interaction. The 62 arms ran as 105 one-GPU jobs on Delta; 163 of their 186 configs are scored, the rest rerunning after Lustre write errors.

The control and the noise. Every arm is read against the round-ten Bliss arm, re-run in this round on the same folds, seeds and inputs. Its growth AUROC is 0.850 and its compound-cold Vanacloig score 0.265. A single seed scatters around its fold's three-seed ensemble by 0.111 on the Vanacloig score and 0.018 on the AUROC; the 63 combinations are the same wells on every fold, so the fold-to-fold spread of a host score, 0.041 for the control, is the floor a host difference has to clear. Everything here is one fold partition.

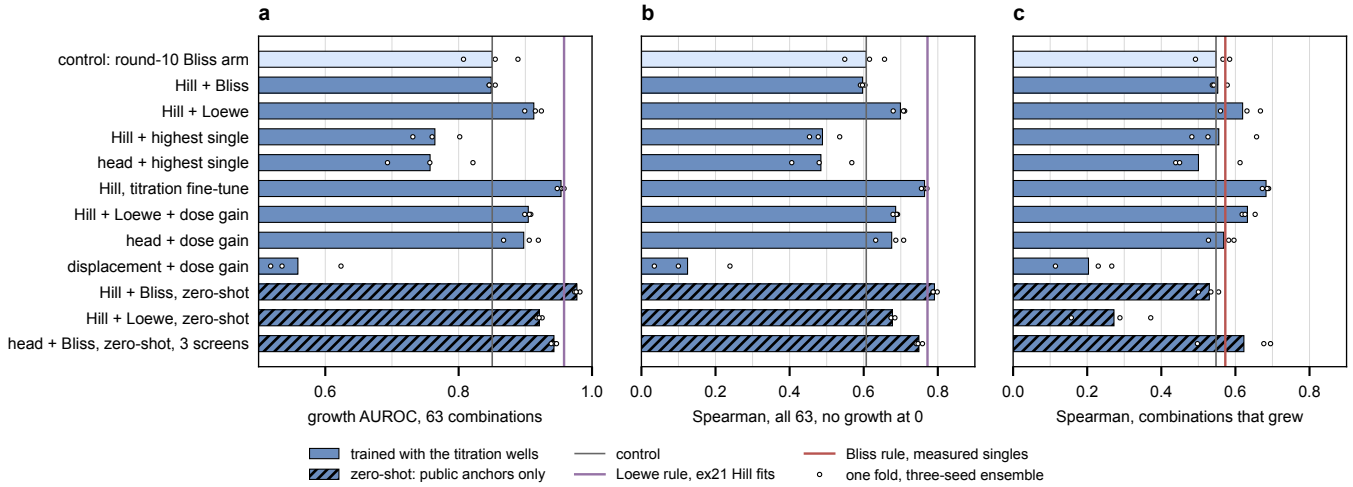


Figure 13: Round eleven on the wet-lab wells. One row per arm, the bar the mean over folds of each fold's three-seed ensemble, the open circles the folds. Hatched arms never saw a titration or a combination well: their only host records are the public IC30 anchors and the compound-free medium. Vertical lines: the control (gray), Loewe additivity from the `ex21` Hill fits (purple), Bliss independence from the measured singles (red). **a**, Growth AUROC over the 63 combinations. **b**, Spearman over the same combinations with no growth read as zero, the ranking a process engineer needs. **c**, Fitness Spearman over the combinations that grew. The displacement arm with the dose gain is the only arm that falls far below the control.

The reading. The Hill form composed by Bliss and trained on the public anchors alone orders the combinations at a growth AUROC of 0.977 (folds 0.974 to 0.982) against Loewe's 0.958, and ranks all of them at a worst-scenario Spearman of 0.791; it is the first arm of the experiment above the Loewe rule, on one partition and three seeds, with single seeds from 0.942 to 0.982. Fitted on the titration, the decoder's rules order as the model-free rules do: Loewe 0.913, Bliss 0.849, the highest single agent 0.764. The Hill fine-tune is the balanced arm (0.954 growth, 0.683 on the combinations that grew, against Bliss's 0.573). Displacing the gene embeddings leaves the public score (0.276 to 0.325) and the host unchanged, and with the dose gain it reads the titration at 0.937 while composition collapses to 0.559. No arm reaches nested ridge's 0.359 on the public side, no dose channel beats none on a held-out public dose (0.258 without one), and the gene-cold split reads zero (-0.001 to 0.007). Figure 14 wires the leading arm.